

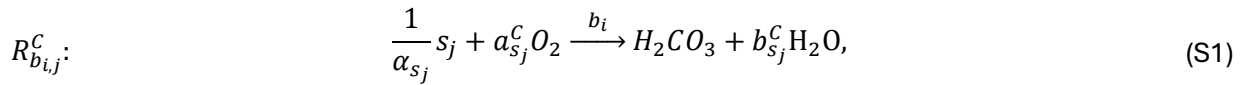
Supplemental Material

Deriving microbial food web structure by maximizing entropy production

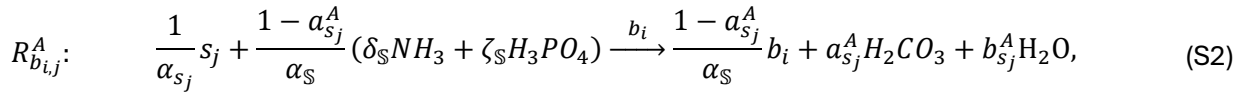
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Bacteria Reactions

Growth of bacteria on substrates is represented by two chemical reactions, one capturing catabolism (energy acquisition) per unit C, given by,



and the other representing anabolism (growth) per unit C, given by,



where s_j is one of n_s substrates that matches the five used in the stable isotope probing (SIP)

experiment (methanol, acetate, ethanol, xylose or glucose) with elemental composition given by

$C_{\alpha_{s_j}} H_{\beta_{s_j}} O_{\gamma_{s_j}}$, b_i is bacteria i , of which there are n_b instances that all have the same elemental

composition given by $C_{\alpha_{\mathbb{S}}} H_{\beta_{\mathbb{S}}} O_{\gamma_{\mathbb{S}}} N_{\delta_{\mathbb{S}}} P_{\zeta_{\mathbb{S}}}$, and $r_{b_{i,j}}^C$ and $r_{b_{i,j}}^A$ designate catabolic and anabolic reaction

rates catalyzed by bacteria b_i utilizing substrate s_j , based on reactions $R_{b_{i,j}}^C$ and $R_{b_{i,j}}^A$ respectively.

The four stoichiometric coefficients, $a_{s_j}^C$, $b_{s_j}^C$, $a_{s_j}^A$ and $b_{s_j}^A$, are readily determined from elemental

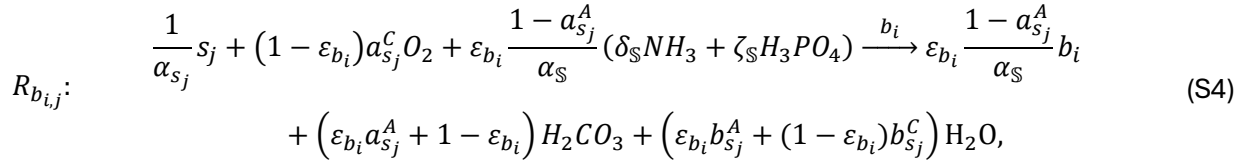
balances around H and O in each reaction. For instance, $a_{s_j}^A$ is given by,

$$a_{s_j}^A = \frac{\alpha_{s_j}(\beta_{\mathbb{S}} - 2\gamma_{\mathbb{S}} - 3\delta_{\mathbb{S}} + 5\zeta_{\mathbb{S}}) - \alpha_{\mathbb{S}}(\beta_{s_j} - 2\gamma_{s_j})}{\alpha_{s_j}(\alpha_{\mathbb{S}} + \beta_{\mathbb{S}} - 2\gamma_{\mathbb{S}} - 3\delta_{\mathbb{S}} + 5\zeta_{\mathbb{S}})}. \quad (S3)$$

The overall reaction representing grown of each bacteria on the n_s substrates is constructed by

multiplying $R_{b_{i,j}}^A$ by ε_{b_i} and adding it to $R_{b_{i,j}}^C$ multiplied by $(1 - \varepsilon_{b_i})$, where ε_{b_i} is the thermodynamic

or growth efficiency parameter, bounded between 0 and 1, that is determined by maximizing entropy production over a specified interval of time. The overall chemical reaction describing bacterial growth is then,



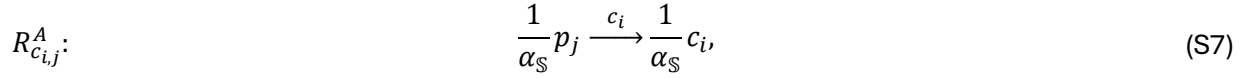
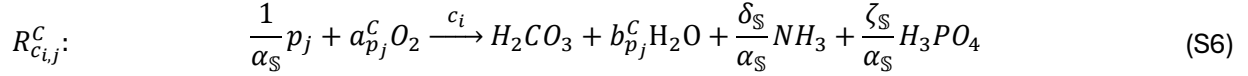
where the Gibbs free energy of reaction for the catabolic, $\Delta G(R_{b_{i,j}}^C)$, and anabolic, $\Delta G(R_{b_{i,j}}^A)$, reactions $R_{b_{i,j}}^C$ and $R_{b_{i,j}}^A$ are calculated from Gibbs free energies of formation obtained from Alberty [1] at the given temperature, pH, and ionic strength, and the concentrations of all chemical species, ρ , via the natural logarithm of the reaction quotient, Q , temperature, and the ideal gas constant (i.e., $\Delta G = \Delta G^\circ + R_k T \ln(Q)$). The overall reaction free energy for $R_{b_{i,j}}$, Eq. (S4), is then given by,

$$\Delta G(R_{b_{i,j}}) = \varepsilon_{b_i} \Delta G(R_{b_{i,j}}^A) + (1 - \varepsilon_{b_i}) \Delta G(R_{b_{i,j}}^C), \quad (S5)$$

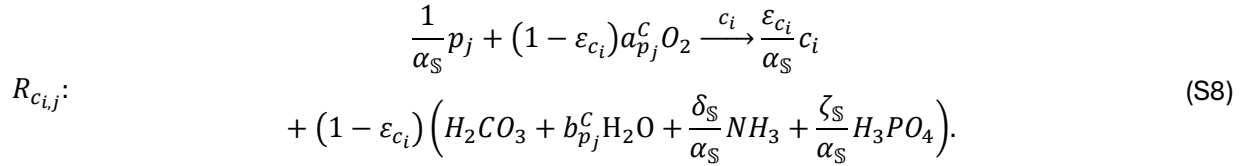
which is strongly controlled by the value of ε_{b_i} . As $\varepsilon_{b_i} \rightarrow 0$, the substrate is completely oxidized, and the free energy (chemical potential energy) is dissipated as heat, resulting in maximum instantaneous entropy production; as $\varepsilon_{b_i} \rightarrow 1$, chemical potential energy is largely conserved, resulting in little to no entropy production (see below).

Consumer Reactions

The reactions of the consumers are handled similarly to that for bacteria. The catabolic and anabolic reactions representing the growth of consumers, c_i , (i.e., protist) on prey, p_j , that includes both bacteria and the consumers themselves (i.e., $p_j = \begin{bmatrix} b_j \\ c_j \end{bmatrix}$, $n_p = n_b + n_c$, see Fig. 1), on a unit carbon bases are given by,



where we have made the simplification that the elemental composition of the consumers, c_i , is the same as bacteria, b_i , and is also given by $C_{\alpha_S} H_{\beta_S} O_{\gamma_S} N_{\delta_S} P_{\zeta_S}$. While this simplification is not necessary for the model, we do not have information regarding compositional differences between prey and consumer, so we have chosen the mathematically cleaner version. The two consumer reactions are also combined via the thermodynamic efficiency parameter, ε_{c_i} , to produce the overall reaction for growth and respiration of consumers,



The equations for Gibbs free energy of reaction, $\Delta G(R_{c_{i,j}})$, are similarly derived to those described above for bacteria.

Reaction Rates

Reaction rates for all bacteria, $r_{b_{i,j}} \in \mathbb{R}^{n_b \times n_s}$, take the general form of,

$$r_{b_{i,j}} = \Phi_{b_{i,j}} B'_{i,j} \rho_{b_i}, \quad (S9)$$

where the matrix $\Phi_b \in \mathbb{R}^{n_b \times n_s}$ is the substrate specific-uptake rate based on an adaptive Monod model [2] (see Eq. (S11) below), $\mathbf{B}'$ is a substrate weighted $\mathbf{B} \in \mathbb{R}^{n_b \times n_s}$ matrix, and ρ_{b_i} is the concentration of b_i (mmol m⁻³). The $\mathbf{B}'$ matrix is given by,

$$\mathbf{B}'_i = \frac{\mathbf{B}_i \odot \boldsymbol{\rho}_s}{\mathbf{B}_i \cdot \boldsymbol{\rho}_s}, \quad (S10)$$

where $\mathbf{B}_i \in \mathbb{R}^{1 \times n_s}$ is the i^{th} row vector of $\mathbf{B}$ and analogously for $\mathbf{B}'_i$, and $\odot$ and $\cdot$ are the elementwise (Hadamard) and dot products, respectively. Rows of the substrate preference matrix, $\mathbf{B}_i$, sum to unity, and determine how much catalyst from bacteria i is allocated to each of the n_s substrates. For example, when $n_s = 5$, bacterial specialists can be represented with all the weight on one substrate, such as $\mathbf{B}_i = [0 \ 1 \ 0 \ 0 \ 0]$, while generalists spread metabolic resources across the 5 substrates, such as $\mathbf{B}_i = [0.2 \ 0.3 \ 0.2 \ 0.1 \ 0.2]$.

The substrate specific-uptake rate by bacteria i using s_j , O_2 , NH_3 and H_3PO_4 based on reaction $R_{b_{i,j}}$ is given by,

$$\Phi_{b_{i,j}} = v^* \varepsilon_{b_i}^2 \left(\frac{\rho_{s_j}}{\rho_{s_j} + \frac{1}{\alpha_{s_j}} \kappa^* \varepsilon_{b_i}^4} \right) \left(\frac{\rho_{O_2}}{\rho_{O_2} + \alpha_{s_j}^C \kappa^* \varepsilon_{b_i}^4} \right) \left(\frac{\rho_{NH_3}}{\rho_{NH_3} + \frac{1}{\alpha_S} \delta_S (1 - \alpha_{s_j}^A) \kappa^* \varepsilon_{b_i}^4} \right) \times \left(\frac{\rho_{H_3PO_4}}{\rho_{H_3PO_4} + \frac{1}{\alpha_S} \zeta_S (1 - \alpha_{s_j}^A) \kappa^* \varepsilon_{b_i}^4} \right) F_T \left(\Delta G \left(R_{b_{i,j}} \right), n_{b_{i,j}}^e \right), \quad (S11)$$

where F_T is the thermodynamic driver described by LaRowe [3] that inhibits the reaction as

$\Delta G \left(R_{b_{i,j}} \right) / n_{b_{i,j}}^e \rightarrow 0$, $n_{b_{i,j}}^e$ is the number of electrons transferred in the catabolic reaction, given by $4\alpha_{s_j}^C$, and v^* and κ^* are fixed constants independent of the substrate or phyla, and equal $350. \text{ d}^{-1}$ and $5000. \text{ mmol m}^{-3}$, respectively.

Reaction rates for consumers follow a similar form to that for bacteria, given by,

$$r_{c_{i,j}} = \Phi_{c_{i,j}} G'_{i,j} \rho_{c_i}, \quad (S12)$$

where $\Phi_c \in \mathbb{R}^{n_c \times (n_b + n_c)}$ is the prey specific-uptake rate, $\rho_c \in \mathbb{R}^{(n_b + n_c) \times 1}$ is the consumer concentrations, and $\mathbf{C}'$ is a prey-weighted $\mathbf{C}$ matrix which determines which prey each consumer eats. The prey-weighted $\mathbf{C}'$ matrix is given by,

$$\mathbf{C}'_i = \frac{\mathbf{C}_i \odot \boldsymbol{\rho}_p}{\mathbf{C}_i \cdot \boldsymbol{\rho}_p}, \quad (\text{S13})$$

where $\mathbf{C}_i \in \mathbb{R}^{1 \times (n_b + n_c)}$ is the i^{th} row vector of $\mathbf{C}$ and analogously for $\mathbf{C}'_i$. The prey specific-uptake rate matrix based on reaction $R_{c_{i,j}}$ is given by,

$$\Phi_{c_{i,j}} = v^* \varepsilon_{c_i}^2 \left(\frac{\rho_{p_j}}{\rho_{p_j} + \frac{1}{\alpha_s} \kappa^* \varepsilon_{c_i}^4} \right) \left(\frac{\rho_{O_2}}{\rho_{O_2} + a_{p_j}^C \kappa^* \varepsilon_{c_i}^4} \right) F_T \left(\Delta G \left(R_{c_{i,j}} \right), n_{c_{i,j}}^e \right), \quad (\text{S14})$$

where $n_{c_{i,j}}^e = 4a_{p_j}^C$.

State Space Model

The food web structure, or connectivity, of a system composed of n_s substrates, n_b bacteria and n_c consumers is described by the two matrices, $\mathbf{B} \in \mathbb{R}^{n_b \times n_s}$ and $\mathbf{C} \in \mathbb{R}^{n_c \times (n_b + n_c)}$, that define which substrates are consumed by each bacteria and which prey each consumer eats, respectively (Fig. 1 in text). To model the experimental chemostats, we use the following state variables: ρ_{s_i} , molar concentration of carbon substrates in the medium; ρ_{b_i} concentrations of bacteria; ρ_{c_i} , concentrations of consumers. To model observed experimental variables, we also include: ρ_{O_2} , concentration of dissolved oxygen; p_{O_2} , partial pressure of oxygen in bioreactor headspace; $\rho_{H_2CO_3}$, concentration of all carbonate species; p_{CO_2} , partial pressure of CO_2 in headspace; $\rho_{H_3PO_4}$, concentration of inorganic phosphate. All concentrations are in mmol m^{-3} , partial pressures are in mbar and nitrogen species are not included as state variables because the experimental system is designed to be P limited. However, ammonium concentration is fixed as a parameter in the model's input file for thermodynamic calculations (a value of $10 \mu\text{M}$ was used). The differential equations governing the state variables, given the reaction rates defined above, are given by,

Food web variables

$$\frac{d\rho_{s_i}}{dt} = \frac{F_L}{V_L}(\rho_{s_i}^f - \rho_{s_i}) - \sum_{j=1}^{n_b} \frac{r_{b_{j,i}}}{\alpha_{s_i}} \quad (\text{S15})$$

$$\frac{d\rho_{b_i}}{dt} = \frac{F_L}{V_L}(\rho_{b_i}^f - \rho_{b_i}) + \sum_{j=1}^{n_s} (1 - a_{s_j}^A) \frac{\varepsilon_{b_i}}{\alpha_{\mathbb{S}}} r_{b_{i,j}} - \sum_{j=1}^{n_c} \frac{r_{c_{j,i}}}{\alpha_{\mathbb{S}}} \quad (\text{S16})$$

$$\frac{d\rho_{c_i}}{dt} = \frac{F_L}{V_L}(\rho_{c_i}^f - \rho_{c_i}) + \sum_{j=1}^{n_p} \frac{\varepsilon_{c_i} r_{c_{i,j}}}{\alpha_{\mathbb{S}}} - \sum_{j=1}^{n_c} \frac{r_{c_{j,i+n_b}}}{\alpha_{\mathbb{S}}} \quad (\text{S17})$$

Gases, inorganic carbon, and phosphate

$$\begin{aligned} \frac{d\rho_{O_2}}{dt} = & \frac{F_L}{V_L}(\rho_{O_2}^f - \rho_{O_2}) + \frac{k_L^{O_2} A}{V_L}(p_{O_2} h_{O_2}(T) - \rho_{O_2}) - \sum_{i=1}^{n_b} \sum_{j=1}^{n_s} (1 - \varepsilon_{b_i}) a_{s_j}^C r_{b_{i,j}} \\ & - \sum_{i=1}^{n_c} \sum_{j=1}^{n_p} (1 - \varepsilon_{c_i}) a_{p_j}^C r_{c_{i,j}} \end{aligned} \quad (\text{S18})$$

$$\frac{dp_{O_2}}{dt} = \frac{F_G}{V_G}(p_{O_2}^f - p_{O_2}) + \frac{k_L^{O_2} A}{V_G} R_{bar} T (p_{O_2} - p_{O_2} h_{O_2}(T)) \quad (\text{S19})$$

$$\begin{aligned} \frac{d\rho_{H_2CO_3}}{dt} = & \frac{F_L}{V_L}(\rho_{H_2CO_3}^f - \rho_{H_2CO_3}) + \frac{k_L^{CO_2} A}{V_L}(p_{CO_2} h_{CO_2}(T) - \rho_{CO_2}) \\ & + \sum_{i=1}^{n_b} \sum_{j=1}^{n_s} (\varepsilon_{b_i} a_{s_j}^A + 1 - \varepsilon_{b_i}) r_{b_{i,j}} + \sum_{i=1}^{n_c} \sum_{j=1}^{n_p} (1 - \varepsilon_{c_i}) r_{c_{i,j}} \end{aligned} \quad (\text{S20})$$

$$\frac{dp_{CO_2}}{dt} = \frac{F_G}{V_G}(p_{CO_2}^f - p_{CO_2}) + \frac{k_L^{CO_2} A}{V_G} R_{bar} T (p_{CO_2} - p_{CO_2} h_{CO_2}(T)) \quad (\text{S21})$$

$$\begin{aligned} \frac{d\rho_{H_3PO_4}}{dt} = & \frac{F_L}{V_L} (\rho_{H_3PO_4}^f - \rho_{H_3PO_4}) - \sum_{i=1}^{n_b} \sum_{j=1}^{n_s} (1 - \alpha_{s_j}^A) \frac{\varepsilon_{b_i} \zeta_{\mathbb{S}}}{\alpha_{\mathbb{S}}} r_{b_{i,j}} \\ & + \sum_{i=1}^{n_c} \sum_{j=1}^{n_p} (1 - \varepsilon_{c_i}) \frac{\zeta_{\mathbb{S}}}{\alpha_{\mathbb{S}}} r_{c_{i,j}} \end{aligned} \quad (S22)$$

where F_L and F_G are volumetric flow rates ($\text{m}^3 \text{d}^{-1}$) of liquid and gas streams into the bioreactor that have concentrations of ρ_i^f and p_i^f for chemical species i , respectively, k_L^i is the liquid-side mass transfer coefficient for species i (aka piston velocity, m d^{-1}), A is the total gas-water interface area including bubbles (m^2) determined experimentally [4], V_L and V_G are the liquid and gas volumes (m^3) in the bioreactor, respectively, and $h_i(T)$ is the Henry's law constant ($\text{mmol m}^{-3} \text{mbar}^{-1}$) for gas i at temperature T [5]. The set of ODE's are integrated using a Blended implicit Method (BiM) [6].

Entropy Production

The instantaneous entropy production from irreversible processes, $\dot{\sigma}$ ($\text{J K}^{-1} \text{d}^{-1}$), occurs as a result of dissipating chemical potential energy stored in substrates, bacteria, and consumers, which is calculated by the product of the reaction free energies, the reaction rate, temperature and system volume. Entropy production from mixing and flow is negligible [7] in these systems and is ignored here. The total entropy production associated with bacterial growth and respiration is given by,

$$\dot{\sigma}_{b_T} = -\frac{V_L}{T} \sum_{i=1}^{n_b} \sum_{j=1}^{n_s} r_{b_{i,j}} \Delta G(R_{b_{i,j}}), \quad (S23)$$

and the total entropy production from consumer growth and respiration is similarly given by,

$$\dot{\sigma}_{c_T} = -\frac{V_L}{T} \sum_{i=1}^{n_c} \sum_{j=1}^{n_p} r_{c_{i,j}} \Delta G(R_{c_{i,j}}), \quad (S24)$$

where T and V_L are temperature and the liquid volume of the bioreactor.

As described in the main text, abiotic systems differ from biological ones in that the latter maximizes entropy production over time, which can be calculated by integrating $\dot{\sigma}_{b_T}$ and $\dot{\sigma}_{c_T}$ over an interval of time, as given by,

$$\sigma|_{t_{InvS}}^{\delta_{MEP}} = \int_{t_{InvS}}^{t_{InvS} + \delta_{MEP}} \dot{\sigma}_{b_T} + \dot{\sigma}_{c_T} d\tau, \quad (S25)$$

where $\sigma|_{t_{InvS}}^{\delta_{MEP}}$ is the entropy production from internal irreversible processes over δ_{MEP} days starting at time t_{InvS} , where t_{InvS} was set to zero for the simulations in this study. Entropy production over the entire simulation time, t_f , is given by $\sigma|_0^{t_f}$.

Entropy production maximization

Food web dynamics, and associated entropy production, depend on three classes of “traits” or control variables, namely, the growth efficiencies of the bacteria and consumers, ϵ_b and ϵ_c , the elements of the bacterial substrate preference matrix, $\mathbf{B}$, and the elements of the consumer prey preference matrix, $\mathbf{C}$. Given the dimensions of these vectors and matrices, the total number of unknowns is $n_u = n_b + n_c + n_b(n_s + n_c) + n_c^2$. To determine the values of the n_u traits, an optimization problem was defined that maximizes entropy production over a specified interval of time while solving the governing set of ODEs, Eqs. (S15 - S22), as given by,

$$J = \max_{\epsilon_b, \epsilon_c, \mathbf{B}, \mathbf{C}} \sigma|_{t_{InvS}}^{\delta_{MEP}} \quad (S26)$$

This optimization problem was solved for different values of δ_{MEP} to examine how the timescale changes the diversity and types of substrates bacteria consume (i.e., generalist versus specialists) and the trophic structure of the consumers.

We used a parallel version of BOBYQA [8], hyperBOB [9], that uses a derivative-free optimization approach to find local optima. Because there is no guarantee of finding the global optimum, we ran hyperBOB on a cluster with 200 CPU cores and selected the best solution from the 200 returned.

Food web metrics

Hill Numbers [10] were used to calculate the substrate diversity metric for bacteria based on the row vectors of the **B** matrix as follows,

$${}^qD_i = \left(\sum_{j=1}^{n_s} \mathbf{B}_{i,j}^q \right)^{1/(1-q)}, \quad (\text{S27})$$

where q was set to 2, so that 2D_i is an arithmetic mean weighted substrate diversity index. The trophic position of a consumer in the food web, T_{c_i} , was calculated from the **C** matrix, which is a sub matrix of the full transfer matrix (Fig. S1), as described by Levin [11], which is given by,

$$T_{c_i} = \sum_{j=1}^{n_p} \left(\mathbf{I} - \begin{bmatrix} \mathbf{0} \\ \mathbf{C} \end{bmatrix} \right)^{-1}_{n_b+i,j} \quad (\text{S28})$$

where the zero matrix, **0**, has the dimensions of $n_b \times n_p$, and the identity matrix, **I**, is $n_p \times n_p$.

All source code (Fortran) for the model runs is available as a Github repo [12].

Supplemental Figures & Tables

Table S1. Nomenclature*

Symbol	Definition
$\mathbf{0}$	Zero matrix ($n_b \times n_p$)
$a_{s_j}^A$	Stoichiometric coefficient for substrate s_j to balance H and O, anabolic reaction (S2)
$a_{p_j}^C$	Stoichiometric coefficient for prey p_j to balance H and O, catabolic reaction (S6)
$a_{s_j}^C$	Stoichiometric coefficient for substrate s_j to balance H and O, catabolic reaction (S1)
A	Air-liquid interface area (m^2)
$b_{p_j}^C$	Stoichiometric coefficient for prey p_j to balance H and O, catabolic reaction (S6)
$b_{s_j}^A$	Stoichiometric coefficient for substrate s_j to balance H and O, anabolic reaction (S2)
$b_{s_j}^C$	Stoichiometric coefficient for substrate s_j to balance H and O, catabolic reaction (S1)
b_i	Bacteria i
$\mathbf{B}$	A matrix of n_b rows and n_s columns that specify the weight bacteria place on uptake of each substrate. Rows must sum to 1. Elements determined by optimization.
$\mathbf{B}'$	Weighted version of $\mathbf{B}$ based on substrate concentrations (S10)
c_i	Consumer i
$\mathbf{C}$	A matrix of n_c rows and n_p columns that specify the weight consumers place on uptake of each prey. Rows must sum to 1. Elements determined by optimization.
$\mathbf{C}'$	Weighted version of $\mathbf{C}$ based on prey concentrations (S13)
qD_i	Substrate diversity metric for bacteria b_i (S27)
F_G	Gas flow rate into reactor ($m^3 d^{-1}$)
F_L	Liquid flow rate of media into reactor ($m^3 d^{-1}$)
F_T	Thermodynamic driver (see LaRowe et al. 2012)
$h_y(T)$	Henry's law constant for gas $y \in [O_2, CO_2]$ ($mmol m^{-3} mbar^{-1}$)
$\mathbf{I}$	Identity matrix ($n_p \times n_p$)
J	Optimization objective function (S26)
k_L^y	Liquid side mass-transfer coefficient for gas $y \in [O_2, CO_2]$ ($m d^{-1}$)
n_b	Number of bacteria modeled

$n_{b_{i,j}}^e$	Number of electrons consumed in reaction $R_{b_{i,j}}^C$
$n_{c_{i,j}}^e$	Number of electrons consumed in reaction $R_{c_{i,j}}^C$
n_c	Number of consumers modeled
n_p	Number of prey modeled ($n_b + n_c$)
n_s	Number of substrates modeled
n_u	Total number of optimization variables ($n_b + n_c + n_b(n_s + n_c) + n_c^2$)
p_j	Prey j ($b_j + c_j$)
p_y	Partial pressure of state variable $y \in [O_2, CO_2]$ (mbar)
p_y^f	Partial pressure of state variable y in feed gas (mbar)
q	Hill number used in (S27). Set to 2.
Q	Reaction quotient (see computer code for details)
$r_{b_{i,j}}$	Rate of reaction $R_{b_{i,j}}$ ($mmol\ m^{-3}\ d^{-1}$)
$r_{c_{i,j}}$	Rate of reaction $R_{c_{i,j}}$ ($mmol\ m^{-3}\ d^{-1}$)
$R_{b_{i,j}}^A$	Reaction stoichiometry of anabolic bacterial reaction
$R_{b_{i,j}}^C$	Reaction stoichiometry of catabolic bacterial reaction
$R_{b_{i,j}}$	Reaction stoichiometry of bacteria b_i consuming substrate s_j
$R_{c_{i,j}}^A$	Reaction stoichiometry of anabolic consumer reaction
$R_{c_{i,j}}^C$	Reaction stoichiometry of catabolic consumer reaction
$R_{c_{i,j}}$	Reaction stoichiometry of consumer c_i consuming prey p_j
R_{bar}	Ideal gas constant ($mbar\ m^3\ mmol^{-1}\ K^{-1}$)
R_{kJ}	Ideal gas constant ($J\ mmol^{-1}\ K^{-1}$)
s_j	Substrate j , where $j \in [m: methanol, a: acetate, e: ethanol, x: xylose, g: glucose]$
t_{InvS}	Start time for entropy production integration (set to 0 in all studies) (d)
t_f	Total simulation time (d)
T	Temperature (K)
T_{c_i}	Trophic position of consumer c_i (S28)
V_G	Gas volume of chemostat (m^3)

V_L	Liquid volume of chemostat (m^3)
α_{s_j}	C atoms in substrate s_j
α_S	C atoms in bacteria and consumers
β_{s_j}	H atoms in substrate s_j
β_S	H atoms in bacteria and consumers
γ_{s_j}	O atoms in substrate s_j
γ_S	O atoms in bacteria and consumers
δ_S	N atoms in bacteria and consumers
δ_{MEP}	Interval of time over which integrated entropy production is maximized (d)
$\Delta G^\circ(R_x)$	Standard free energy of reaction R_x at specified T , pH, and ionic strength ($J\ mmol^{-1}$)
$\Delta G(R_x)$	Free energy of reaction R_x ($J\ mmol^{-1}$)
Δt	Interval of time (d)
ε_{b_i}	Growth efficiency of bacteria b_i , determined by optimization
ε_{c_i}	Growth efficiency of consumer c_i , determined by optimization
ζ_S	P atoms in bacteria and consumers
κ^*	Universal half saturation constant ($5000\ mmol\ m^{-3}$)
ν^*	Universal specific uptake rate ($350\ d^{-1}$)
$\Phi_{b_{i,j}}$	Substrate specific uptake rate of substrate s_j by bacteria b_i (S11) (d^{-1})
$\Phi_{c_{i,j}}$	Prey specific uptake rate of prey p_j by consumer c_i (S14) (d^{-1})
ρ_{b_i}	Concentration of bacteria b_i ($mmol\ m^{-3}$)
ρ_{c_i}	Concentration of consumer c_i ($mmol\ m^{-3}$)
ρ_{p_i}	Concentration of prey p_i ($mmol\ m^{-3}$)
$\rho_{\mathbf{x}}$	Concentration of state variable $\mathbf{x} \in [\mathbf{s}, \mathbf{b}, \mathbf{c}, O_2, H_2CO_3, H_3PO_4]$ ($mmol\ m^{-3}$)
$\rho_{\mathbf{x}}^f$	Concentration of state variable $\mathbf{x}$ in media feed ($mmol\ m^{-3}$)
$\dot{\sigma}(t)$	Entropy production from irreversible processes ($J\ d^{-1}\ K^{-1}$)
$\dot{\sigma}_{b_T}$	Entropy production from irreversible processes associated with bacterial growth ($J\ d^{-1}\ K^{-1}$)

$\dot{\sigma}_{c_T}$ Entropy production from irreversible processes associated with consumer growth
($J d^{-1} K^{-1}$)

$\sigma|_{t_{InvS}}^{\delta_{MEP}}$ Integrated entropy production from t_{InvS} to $t_{InvS} + \delta_{MEP}$ ($J K^{-1}$)

*See *.inp files at end of supplemental for parameter values.

		Source																			
		Glucose	Acetate	Ethanol	Methanol	Xylose	b1	b2	b3	b4	b5	b6	b7	b8	b9	b10	g1	g2	g3	g4	g5
Consumes	Glucose	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Acetate	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Ethanol	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Methanol	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Xylose	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b1	0.36	0.00	0.60	0.00	0.04	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b2	0.42	0.11	0.28	0.04	0.16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b3	0.29	0.00	0.59	0.00	0.12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b4	0.10	0.00	0.62	0.00	0.29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b5	0.00	0.14	0.32	0.54	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b6	0.00	0.15	0.10	0.85	0.00	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b7	0.36	0.20	0.00	0.00	0.43	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b8	0.21	0.00	0.53	0.00	0.26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b9	0.20	0.02	0.24	0.34	0.20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b10	0.13	0.00	0.61	0.00	0.26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	g1	0	0	0	0	0	0.30	0.01	0.00	0.00	0.00	0.00	0.01	0.28	0.01	0.00	0.01	0.06	0.32	0.00	0.00
	g2	0	0	0	0	0	0.10	0.00	0.03	0.00	0.23	0.22	0.00	0.00	0.00	0.02	0.00	0.00	0.22	0.06	0.06
	g3	0	0	0	0	0	0.12	0.09	0.01	0.12	0.05	0.12	0.01	0.06	0.12	0.03	0.09	0.05	0.00	0.01	0.11
	g4	0	0	0	0	0	0.00	0.00	0.03	0.00	0.04	0.07	0.00	0.00	0.00	0.02	0.00	0.00	0.84	0.00	0.00
	g5	0	0	0	0	0	0.00	0.00	0.00	0.23	0.28	0.28	0.00	0.00	0.00	0.09	0.00	0.00	0.11	0.00	0.00

Fig. S1. Full transfer matrix for a system of 5 substrates, 10 bacteria, and 5 consumers. In the model implemented for this study, it was assumed that there were no cross-feeding, bacterial predators or pathogens, nor mixotrophs. The diagonal purple line on the right side of the **C** matrix highlights cannibalism. The identity matrix, **I**, in the upper left of the transfer matrix signifies that the substrate added went to that substrate. In Levine's [11] transition matrix, his sub matrix **R** equals $\begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix}$ and **Q** equals $\begin{bmatrix} \mathbf{0} \\ \mathbf{C} \end{bmatrix}$.

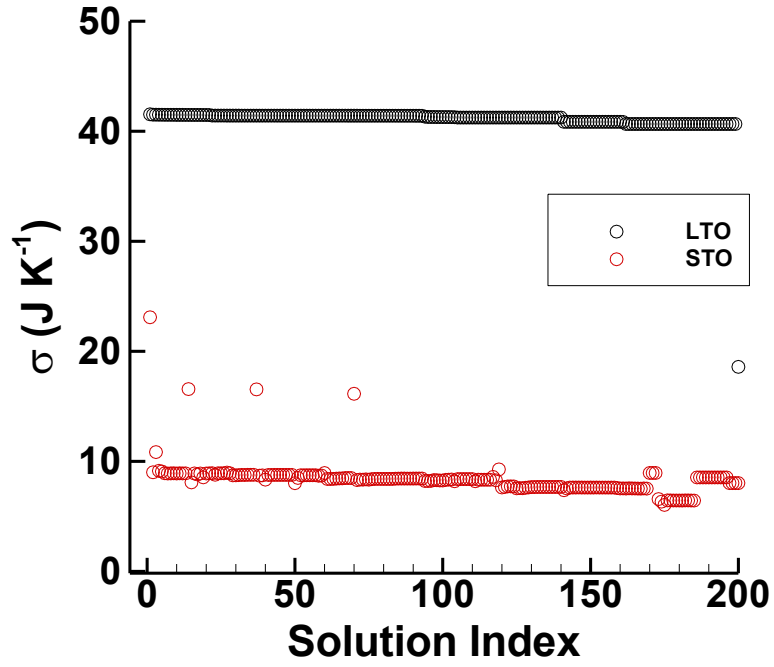


Fig. S2. Integrated entropy production, $\sigma|_0^{t_f}$, from Eq. (S25) over the 53 days of simulation for the LTO (black circles) and STO (red circles) solutions for all 200 local optima found. Solution Index ranking is based on integrated entropy production from best (index 1) to worst (index 200). Note, ranking for the STO solution is based on entropy produced in the first 2 days, not over 53 days.

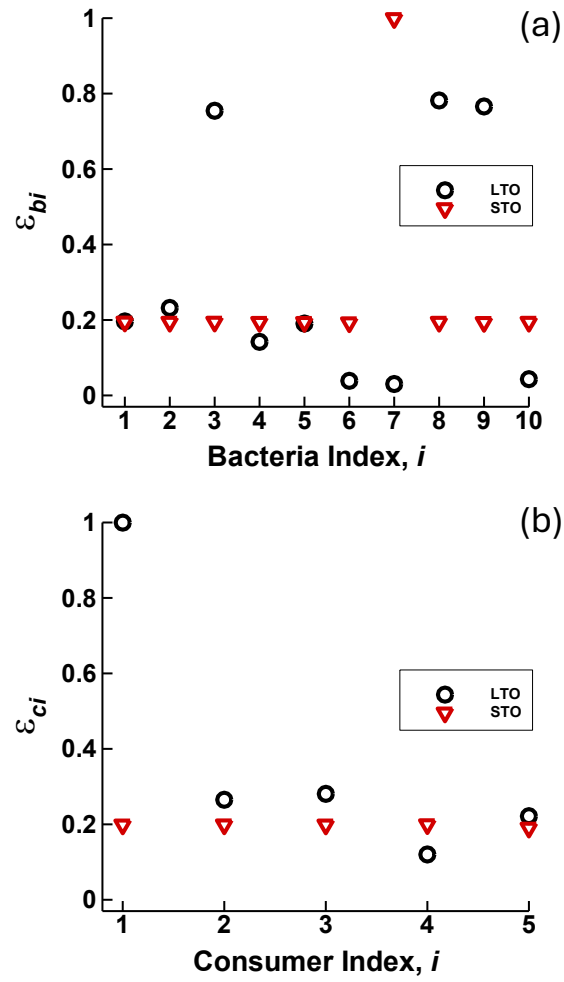


Fig. S3. Growth efficiencies of (a) bacteria, ε_{bi} , and (b) consumers, ε_{ci} , for the top ranked short-term optimization (STO) solution (red triangles) and the long-term optimization (LTO) solution (black circles)

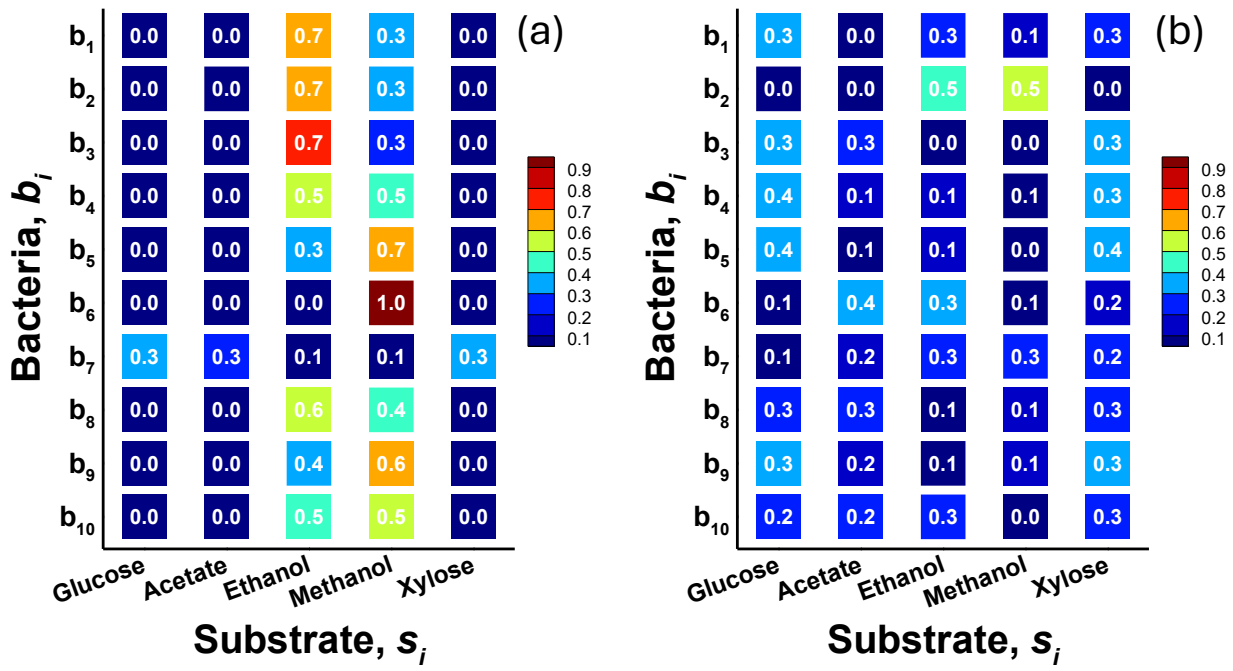


Fig. S4. Substrate preference matrix of bacteria, $\mathbf{B}$, for the top-ranked (a) short- term optimization (STO) solution and (b) long-term optimization (LTO) solution.

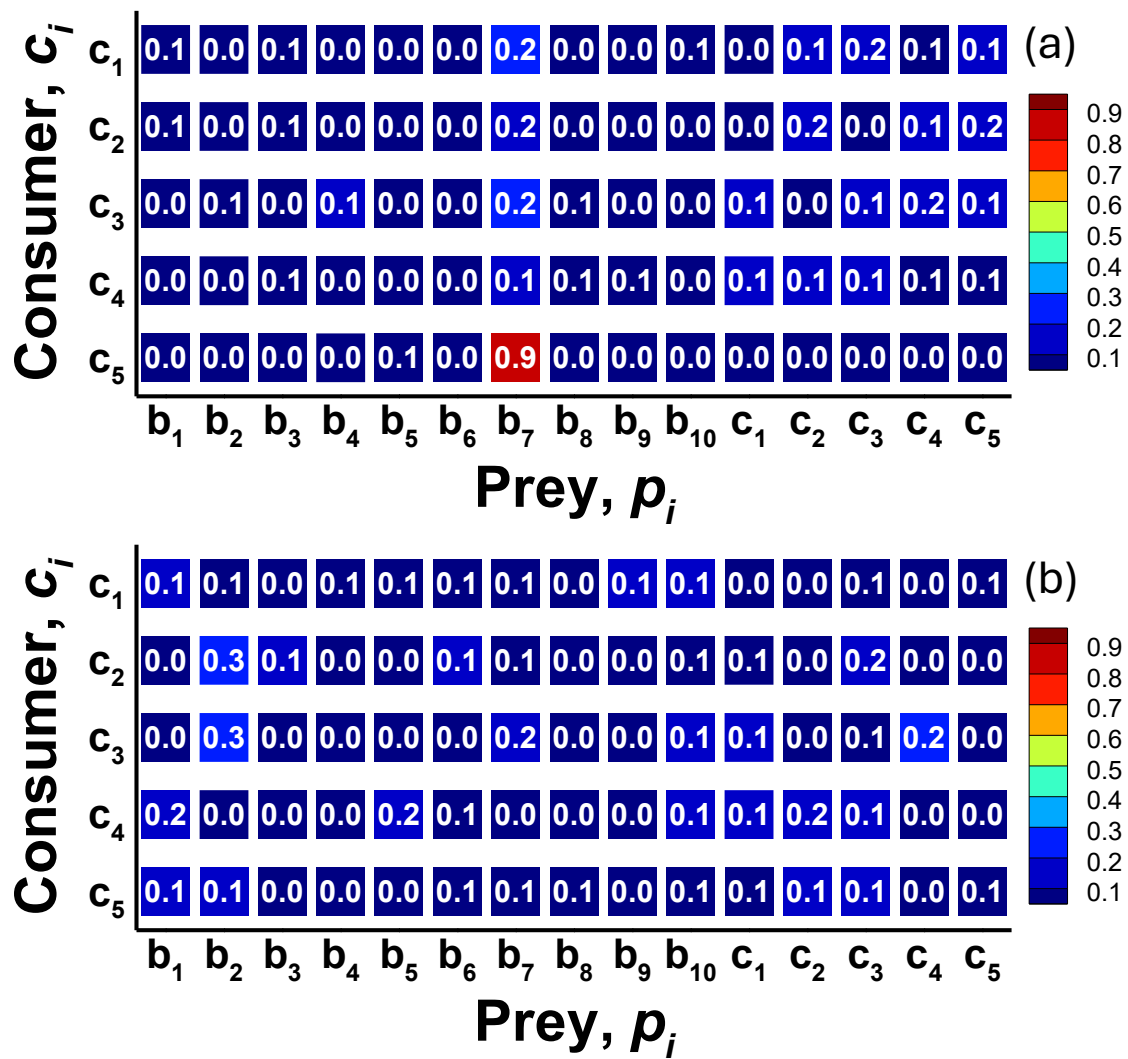


Fig. S5. Prey preference matrix of consumers, C , for the top-ranked (a) short-term optimization (STO) solution and (b) long-term optimization (LTO) solution.

Input files for the Short-Term Optimization (STO) and Long-Term Optimization (LTO) simulations

Run43, STO simulation, *.inp input file:

```
! 11-Apr-2025
!   This is Run20, but using V2.2d instead of V2.2c
!   Darwin_ChemoStat_V2.2d.f90
!       Now effective trophic position of the consumers is calculated.
!   2 day optimization window
!   1 uM P
!   D = 0.1 1/d (F_L = 0.0003 )

&dimens
! problem dimensions
nSub = 5 ! number of substrates
nBac = 10 ! number of bacteria
nGrz = 5 ! number of consumers
/

&parameters
! All parameters used by problem
absZero = 1.d-8 ! prevents division by zero

kappa = 5000. ! (mmole/m3)
nuStar = 350. ! (1/d)
delPsi = 0.1 !0.240 ! LaRowe's thermo driver. Cell membrane potential (V)

alf_Bio = 1.000 ! C elemental composition of phytoplankton, unit carbon based. (use
Battley1998 for yeast)
bet_Bio = 1.613 ! H
gam_Bio = 0.557 ! O
del_Bio = 0.158 ! N
zet_Bio = 0.012 ! P

T_K = 293.15 ! temperature (K)
pH = 6.6 ! pH
is = 0.07 ! ionic strength (M) Can use the formula: 0.72*PSU/35.0 (assume 3.4 PSU
here)

nh3_hold = 10.0 ! Concentration (mmol/m3) of NH3 used in thermo calculations

kL_o2 = 1.5 ! piston velocity for O2 (m/d)
kL_co2 = 1.3 ! piston velocity for CO2 (m/d) Should use Schmidt no. to relate these,
do it later
area_GL = 0.041 ! area of gas-liquid interface (m2) (MC's are about 9" in diameter)
V_L = 0.003 ! volume of liquid in chemostat (m3)
F_L = 0.0003 ! liquid flow rate (m3/d)
V_G = 0.00474 ! volume of gas headspace (m3)
F_G = 0.01545 ! gas flow rate (m3/d) (This is 10 sccm at 273 K, or 10.73 mL/min at 20
C)

! substrate names
subNames(1) = 'glucose'
subNames(2) = 'acetate'
subNames(3) = 'ethanol'
subNames(4) = 'methanol'
subNames(5) = 'xylose'

! substrate CHO composition
CHO(1,:) = 6, 12, 6
CHO(2,:) = 2, 4, 2
CHO(3,:) = 2, 6, 1
```

```

CHO(4,:) = 1, 4, 1
CHO(5,:) = 5, 10, 5

! initial conditions for all bi and gi (mmol/m3)
! The chemostat feed is set to these initial conditions, except no bi or gi in it.
t0 = 0. ! start time. (d)
tDays = 53. ! number of days to run
t0_mep = 0. ! the time (day) that entropy production maximization begins
tDays_mep = 2. ! the number of days over which entropy production is maximized
bi_ini = 0.1
bi_f = 0.0 ! No bacteria in feed
gi_ini = 0.1
gi_f = 0.0 ! No grazers in feed.
! substrates (mmol/m3)
sj_f(1) = 50.0 ! glucose
sj_f(2) = 167.5 ! acetate
sj_f(3) = 108.3 ! ethanol
sj_f(4) = 206.1 ! methanol
sj_f(5) = 59.69 ! xylose
! Chemistry (mmol/m3 and mbar)
chm_f(1) = 250 ! oxygen
chm_f(2) = 212.7 ! O2 pressure (mbar)
chm_f(3) = 210 ! DIC (this one needs some thought, but currently 0.1 of seawater)
chm_f(4) = 0.405 ! pCO2 (mbar)
chm_f(5) = 1. ! phosphate, if using phosphate buffer.
/

&hyperBOBparams
! hyperBOB parameters
optimize = .true. ! if set to false then traits values are randomly set and a
simulation is run. Mostly for testing
rhobeg = 0.49 ! initial and final values of a trust region radius
rhoend = 0.0001
iprint = 0 ! controls amount of printing (0, 1, 2 or 3)
maxfun = 200000 ! maximum number of calls to CALFUN
fcnUpdate = 100 ! After every fcnUpdate PDE integration, info is printed out.
seed_bobyqa = 17 ! used by hyperBOB for random hypercube. All MPI processes use same
value
/

&tecIOparams
! TecIO parameters for output
nIOpts = 1000 ! number of time series points to store in a single zone
debug_IO = 0 ! 1 to debug, 0 no debug.
flushData = .false. ! If true, data is flushed to files at each time output, but may
slow down execution.
/

&BiMparams
! parameters for BiM integrator
rtol = 1.d-6 ! relative tolerance
atol = 1.d-8 ! absolute tolerance
hmax_BiM = 0.1 ! maximum step size. = (tend-t0)/8 if set to 0
maxstep_BiM = 0 ! maximum number of steps. Default 100000 if set to zero
maxattempts = 10 ! maximum number of times to retry integration
ompThreads = 1 ! Number of OMP threads to use for numerical Jacobian calculations
minCompFac = 100000.0 ! if a ODE solution takes longer than (tf-t0)/minCompFac (days),
then it is terminated.
/

```

Run44, LTO simulation, *.inp file:

Same *.inp file as above, except:

```
tDays_mep = 53. ! the number of days over which entropy production is maximized
```

References

- [1] Alberty, R.A. 2003 *Thermodynamics of biochemical reactions*. Hoboken, NJ, Wiley & Sons.
- [2] Vallino, J.J. & Huber, J.A. 2018 Using Maximum Entropy Production to Describe Microbial Biogeochemistry Over Time and Space in a Meromictic Pond. *Frontiers in Environmental Science* **6**, 1-22. (doi:10.3389/fenvs.2018.00100).
- [3] LaRowe, D.E., Dale, A.W., Amend, J.P. & Van Cappellen, P. 2012 Thermodynamic limitations on microbially catalyzed reaction rates. *Geochimica et Cosmochimica Acta* **90**, 96-109. (doi:10.1016/j.gca.2012.05.011).
- [4] Bulseco, A.N., Yang, W., Huber, J.A. & Vallino, J.J. 2024 Growth characteristics of natural microbial populations are skewed towards bacteria with low specific growth rates. *bioRxiv*, 2024.2004.2024.590925. (doi:10.1101/2024.04.24.590925).
- [5] Sander, R. 2023 Compilation of Henry's law constants (version 5.0.0) for water as solvent. *Atmos. Chem. Phys.* **23**, 10901-12440. (doi:10.5194/acp-23-10901-2023).
- [6] Brugnano, L. & Magherini, C. 2004 The BiM code for the numerical solution of ODEs. *Journal of Computational and Applied Mathematics* **164-165**, 145-158. (doi:10.1016/j.cam.2003.09.004).
- [7] Vallino, J.J. 2011 Differences and implications in biogeochemistry from maximizing entropy production locally versus globally. *Earth Syst. Dynam.* **2**, 69-85. (doi:10.5194/esd-2-69-2011).
- [8] Powell, M.J. 2009 The BOBYQA algorithm for bound constrained optimization without derivatives. (pp. 26-46. Cambridge, University of Cambridge.
- [9] Vallino, J.J. 2020 hyperBOB: Optimization using MPI parallel Latin hypercube sampling and BOBYQA. <https://github.com/maxEntropyProd/hyperBOB> (doi:10.5281/zenodo.3978688).
- [10] Tuomisto, H. 2010 A diversity of beta diversities: straightening up a concept gone awry. Part 1. Defining beta diversity as a function of alpha and gamma diversity. *Ecography* **33**, 2-22. (doi:10.1111/j.1600-0587.2009.05880.x).

[11] Levine, S. 1980 Several measures of trophic structure applicable to complex food webs.

Journal of Theoretical Biology **83**, 195-207. (doi:10.1016/0022-5193(80)90288-X).

[12] Vallino, J.J. 2025 MEP-Chemostat model. <https://github.com/maxEntropyProd/MEP-Chemostat>

(doi:10.5281/zenodo.15215254).